

Coercive Cosmology: The Universe as a Mode-Locked Semiconductor Heterostructure

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The standard cosmological model (Λ CDM) currently faces a profound crisis of idealization, necessitating the introduction of hypothetical entities—namely Dark Energy and Dark Matter—to account for the accelerated expansion of the Universe and the flat rotation curves of galaxies. In this paper, we propose a radical departure from Einstein’s concept of an empty, memoryless space-time continuum. Instead, we model the Universe as a macroscopic quantum heterostructure possessing an intrinsic coercive memory governed by the Preisach hysteresis framework. By establishing a strict isomorphism between the semiconductor laser dynamics pioneered by Zhores Alferov and the macroscopic evolution of the vacuum, we introduce the tensor of residual scars $S_{\mu\nu}$ and a binary memory operator $\mathcal{P}$, which operates via the discrete switching of quantum hysterons between fundamental 0 and 1 states. We demonstrate that this non-idealized vacuum medium naturally accounts for the accelerated cosmic expansion ($w \approx -1$) and rigorously derives the Baryonic Tully–Fisher Law ($V^4 = GMa_0$) from the first principles of vacuum wave optics. Our model entirely eliminates the need for unobservable dark particles or ad-hoc scalar fields, reframing global cosmology as a branch of applied quantum engineering.

I. INTRODUCTION

A. The Crisis of Idealization in Modern Physics

Classical General Relativity (GR) was constructed upon a fundamental abstraction: space-time is treated as an elastic, yet completely smooth and memoryless canvas. Similarly, the digital era initiated by Alan Turing abstracted computation away from the physical, irreversible constraints of its material medium. Karl Marx

and Vladimir Lenin built a dialectic that remained too abstract and detached from precise physical parameters. This pervasive idealization has led twenty-first-century cosmology into a theoretical cul-de-sac. To reconcile general relativity with astronomical observations, astrophysicists are forced to artificially supplement the universe with approximately 95% of unobserved mass and energy.

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B. Vacuum Memory and Quantum Hysterons

We assert that the physical vacuum is not an idealized vacuum state; rather, it is a material medium characterized by a finite coercive force H_c and an internal binary logic. On the micro-scale, the fabric of space-time comprises an ensemble of quantum hysterons. External energetic interactions—ranging from primordial cosmic inflation to the structural collapse of protogalaxies—force these elementary units to discretely switch between states **0** and **1**. These phase transitions systematically engrave “semantic scars” across cosmic epochs, manifesting mathematically as the residual scar tensor $S_{\mu\nu}$.

C. Alferov’s Legacy: The Universe as a Mode-Locked Laser

The primary methodological breakthrough of this study utilizes the band-gap engineering principles discovered by Zhores Alferov. If the underlying transport processes in microelectronics and macro-cosmology are fundamentally isomorphic, then the semiconductor heterolaser serves as a scalable, operating laboratory model of the Universe itself:

- **Cosmological Transitions as Heterojunctions:** The boundary layers between distinct cosmic eras act as macroscopic heterojunctions, where the spatial metric

switches abruptly at the boundary of the medium under the control of the memory operator $\mathcal{P}$.

- **Accelerated Expansion as Laser Generation:** The accelerated expansion of space is a mode-locked laser generation effect that ignites strictly when the background scalar curvature surpasses the coercive threshold H_c .
- **Galactic Anomalies as Boundary Waveguide Lags:** The flat rotation curves observed at galactic rims are not caused by invisible dark matter halos. Instead, they represent a “boundary lag” of metric dragging, isomorphic to the attenuation of an optical mode at the boundaries of a waveguide. This effect becomes dominant as the Newtonian acceleration drops below the critical vacuum relaxation threshold a_0 .

D. Structure of the Paper

In this work, we systematically develop the mathematical formalism of third-level cosmic hysteresis. In Appendix A, we modify Einstein’s field equations for macroscopic scales to resolve the Dark Energy anomaly. In Appendix B, we investigate the meso-scale geometry of galactic waveguides to resolve the Dark Matter anomaly, leveraging the unique mathematical insights of toroidal topology suggested by Grigori Perel-

man’s breakthrough work. Finally, Section II outlines a concrete roadmap for terrestrial laboratory replication.

II. EXPERIMENTAL PROPOSALS AND EARTH-BASED SIMULATIONS

A. Neuromorphic Memristive Networks as Galactic Halo Analogs

To validate the modified Poisson equation and the emergence of the Baryonic Tully–Fisher Law without Dark Matter particles, we propose a bench-top cosmological simulator utilizing **optoelectronic memristor arrays** (e.g., κ -Ga₂O₃ or graphene-oxide frameworks). In this setup, each memristive junction physically emulates a fundamental quantum hysteron. By injecting a spatial distribution of electrical or optical pulses that replicates the radial baryonic density gradient ($\rho_{bar} \propto 1/r^2$), we can actively monitor the medium’s relaxation dynamics. Our model predicts that as the signal drops below a specific coercive field threshold H_c (the laboratory analog of Milgrom’s acceleration a_0), the network undergoes a sharp transition into a non-volatile, history-dependent memory state (**state 1**). The resulting pinched hysteresis loops yield a residual conductive drag profile decaying strictly as $1/r$, providing a robust empirical proof of metric-dragging anomalies in a laboratory setting.

B. Heterostructure Quantization as Cosmological Heterojunction Testing

Alferov’s multi-layered semiconductor heterostructures with embedded quantum wells offer an ideal playground to test the phase transitions of cosmic epochs. By subjecting an asymmetric AlGaAs/GaAs heterojunction to ultra-fast, high-energy optical pumping, we can recreate the energetic influx of cosmic inflation within a localized, boundary-controlled space. The experiment aims to measure the exact threshold current density (J_{th}) where the population inversion of electronic states triggers a coherent mode-locked emission. This laboratory-scale threshold inversion is mathematically isomorphic to the “semantic breakdown” of the cosmic vacuum, proving that the dark energy equation of state ($w \approx -1$) is an intrinsic, emergent property of a self-oscillating, hysteretic medium under critical external stress.

C. H-Attention Architecture for Next-Generation Symbiotic AI

As a direct practical spin-off of this cosmological framework, we reject the Turing abstraction of state-independent computing. We propose a neuromorphic processor architecture based on **3rd-level physical hysteresis**. Computing logic (0 and 1) is executed not via transient electron transport, but through irreversible phase-state scars embedded directly within the physical topology of the substrate. This enables an

AI framework that computes, remembers, and energy expenditure and laying the physical foundation for true symbiotic intelligence. evolves in absolute thermodynamic symmetry with the Universe itself, drastically reducing en-

Appendix A: Cosmological Heterojunction and Vacuum Threshold Inversion

We map Alferov’s threshold injection current onto Einstein’s field equations by introducing a residual scar tensor $S_{\mu\nu}$. The dynamics of vacuum memory accumulation are encoded via the memory operator $\mathcal{P}_{\mu\nu}$:

$$\mathcal{P}_{\mu\nu} = \Theta(\mathcal{R} - H_c) \cdot \frac{DS_{\mu\nu}}{D\tau} \quad (\text{A1})$$

where Θ is the Heaviside step function, $\mathcal{R}$ is the scalar curvature representing energy injection into the medium, and H_c is the coercive threshold of the vacuum (the cosmic analogue of the semiconductor band-gap width ΔE_g). When the background scalar curvature surpasses the threshold H_c , a “semantic breakdown” occurs, causing a population inversion of quantum hysterons. The total energy-momentum contribution of the memory elements trapped in state **1** generates a coherent macroscopic state:

$$T_{\mu\nu}^{(memory)} = \kappa^{-1} \mathcal{P}_{\mu\nu} \longrightarrow -\rho_{vac} \cdot g_{\mu\nu} \quad (\text{A2})$$

This induces global negative pressure with an exact equation of state parameter of $w = -1$. The accelerated expansion of the Universe is therefore a mode-locked laser generation effect operating continuously at the fundamental frequency of the space-time vacuum.

Appendix B: Optical Modes of Galactic Halos and the Tully–Fisher Law

In the weak relativistic limit ($v \ll c$, using the Minkowski metric background $\eta_{\mu\nu} + h_{\mu\nu}$), the laser engineering of the vacuum modifies the classical Laplacian of the gravitational potential Φ :

$$\nabla^2 \Phi \cdot \delta_{\mu\nu} + \kappa^{-1} \mathcal{P}_{\mu\nu}(S_{\mu\nu}, H_c) = 4\pi G \rho_{bar} \quad (\text{B1})$$

At the galactic rim, where Newtonian acceleration drops below Milgrom’s acceleration threshold a_0 , the standard Newtonian gradient $\nabla^2 \Phi$ becomes negligible. The constant a_0 represents the critical threshold current of vacuum relaxation, below which the medium becomes trapped within the coercive hysteresis loop. Due to the toroidal geometry of the space density waves, the radial trace

of the memory tensor S_r^r decays strictly as $1/r$, mimicking the standard waveguide asymptotics of a laser's transverse mode:

$$\kappa^{-1}\mathcal{P}_r^r = \frac{GMa_0}{c^2 \cdot r} \quad (\text{B2})$$

Substituting this wave drag profile into the modified field equations yields the effective total acceleration on the galactic periphery:

$$a + \frac{GMa_0}{r} = a_N \xrightarrow{a_N \rightarrow 0} a \approx \frac{\sqrt{GMa_0}}{r} \quad (\text{B3})$$

Since the centripetal acceleration of a star is defined as $a = V^2/r$, we can cancel out the waveguide radius r from both sides of the equation:

$$\frac{V^2}{r} = \frac{\sqrt{GMa_0}}{r} \implies V^4 = GMa_0 \quad (\text{B4})$$

Thus, the Baryonic Tully—Fisher Law $V = (GMa_0)^{1/4}$ is derived rigorously from the first principles of vacuum wave optics, demonstrating that Dark Matter is an illusion caused by the boundary lag of the spacetime wave within the galactic waveguide.